Solar PV System



Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: A complete unit of Solar photovoltaic system; A complete unit of Solar photovoltaic system model of solar photovoltaic power plant; Solar power plant

What it is used for

A solar photovoltaic system converts sunlight into electrical energy and delivers it to approved loads, storage, or the grid through compatible equipment. Correct inspection, safe isolation, and regular performance monitoring help maintain system efficiency and reduce electrical risks.

Main parts

- Solar PV modules convert sunlight into DC electricity.
- Mounting structure supports the modules at the designed angle and position.
- DC cables and connectors carry electricity from the modules.
- Combiner box combines strings and may contain protective devices.
- Solar inverter converts DC electricity into AC electricity.
- AC distribution board routes power through suitable protective devices.
- Earthing and bonding components provide protective electrical paths.
- Monitoring equipment measures generation and system status; batteries may be included in hybrid systems.

Before you start

1. Only trained and authorised personnel should install, inspect, or service electrical parts of the system.
2. Read the system documentation and identify the approved layout and shutdown procedure.
3. Inspect modules, mounting structures, cables, connectors, and visible enclosures for damage.
4. Check that cables, protective devices, and inverter ratings match the approved design.
5. Confirm that the work area is safe, dry where required, and free from obstructions.
6. Identify all energy sources, including PV strings, batteries, and grid connections, before isolation.

How to use

1. Switch off and isolate the system according to the manufacturer's and site procedures before electrical work.
2. Check mounting hardware and module condition from a safe, authorised access position.
3. Inspect visible DC cables, connectors, combiner boxes, and AC distribution equipment for damage.
4. Verify that earthing, bonding, cable routing, and protective devices match the approved design.
5. Restore connections and covers only after the authorised inspection or maintenance is complete.
6. Restore power in the specified sequence and check the inverter and monitoring interface.
7. Confirm that the system starts normally and no unexpected fault indicators appear.

8. Record energy generation, system status, and relevant voltage or current readings where provided.
9. Compare performance with expected conditions and investigate unusual drops or repeated alarms.
10. Report defects and arrange qualified repair before operating equipment with unresolved safety faults.

⚠ Safety

- PV modules can generate DC voltage whenever illuminated; switching off the inverter does not make every conductor safe.
- Never disconnect PV connectors under load unless the equipment procedure explicitly permits it.
- Do not walk on modules or work on roofs without approved fall protection and access arrangements.
- Avoid electrical work in wet conditions unless the approved procedure specifically permits it.
- Use suitable PPE and rated test instruments; never bypass protective devices or earthing.
- Only qualified personnel should work on grid connections, battery circuits, or damaged high-voltage components.

Care & storage

- Keep modules free of excessive dust and shading using approved cleaning methods.
- Inspect mounting hardware, cable supports, connectors, and enclosures at scheduled intervals.
- Review inverter alarms and generation records for abnormal changes.
- Arrange authorised servicing after storms, physical damage, insulation faults, or repeated system trips.

Common mistakes to avoid

- Assuming the PV array is safe because the inverter has been switched off.
- Using incompatible connectors or mixing connector types without approval.
- Ignoring damaged cables, loose mounting hardware, or repeated inverter alarms.
- Cleaning modules or servicing electrical parts without the required isolation and safety precautions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.